AGGRESSION FACTORS IN DOGS

MANYDOGS_2024 SURVEY

DATA606 PROJECT

DECEMBER 3, 2025

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ABSTRACT

This project investigates the factors that contribute to aggression in dogs using data collected from an international survey of dog guardians called the Many Dogs Project[1]. The dependent variable, dog aggression, was derived from a subset of 27 aggression question responses, with summary ratings calculated to represent overall aggression score for each respondent. The dataset includes multiple independent variables, mostly categorical, encompassing aspects of the dog's home environment, demographic characteristics, and "origin" (e.g., shelter, breeder, etc.).

A multiple regression analysis was conducted to examine the relationship between these predictors and aggression. The final model explained a moderate proportion of dog aggression (adjusted R^2 of ~0.2), leaving room for more explanatory variables to be captured in the future. Among the most influential predictors were elements of (1) the dog's home environment (e.g., dogs residing in private homes vs. group settings and the number of other dogs in the home) and (2) the dog's background (e.g., origin & breed).

Regression assumptions, including linearity, homoscedasticity, normality of residuals, and absence of multicollinearity, were assessed and found to be adequately met. The model has limitations in explanatory power and should the opportunity arise, it would be interesting to incorporate more quantitative variables.

Overall, this project provides evidence that both environmental and origin-related factors are predictors of aggression in dogs. Other interesting findings were that (i) sex and urban/rural settings did not have explanatory power and (ii) shelter dogs increased aggression in the model while rescue dogs decreased aggression score. These insights can inform guardians, trainers, and policymakers in developing strategies to manage and prevent aggressive behavior.

1. ManyDogs Project, Espinosa, J., Hare, E., Alberghina, D., Perez Valverde, B, & Stevens, J.R. (2024). Data for ManyDogs 1. *Journal of Open Psychology Data*. doi:10.5334/jopd.109

THIS IS NOLA

OVERVIEW – BACKGROUND DATA

- The data for this project is collected from the **Many Dogs Project**[1].
- Methodology:
  - Dog guardians recruited through research site databases and social media.
  - The guardians would complete an online survey of basic demographics & questions regarding dog temperament and behaviors. There is a second part of the research on site where the dog completes a series of behavioral tasks involving food rewards gauging their interest and behavior with and with-out ostensive communication.
  - This study was conducted over 13 months between **January 2022 to January 2023**.
- There are 704 cases (studies) collected from 20 research sites across eight countries. After filtering for incomplete surveys/NAs, we are left with **155** observations which will be used.
- Research is widely used – e.g., main study looked at how ostensive communication can influence dog behavior.

1. ManyDogs Project, Espinosa, J., Hare, E., Alberghina, D., Perez Valverde, B, & Stevens, J.R. (2024). Data for ManyDogs 1. *Journal of Open Psychology Data*. doi:10.5334/jopd.109

OVERVIEW – DEPENDENT VARIABLE

- Dependent Variable: **Aggression**
  - 27 individual aggression rating questions like "When approached directly by an unfamiliar adult while being walked/exercised on a leash"
  - MC possible responses:
    - No aggression
    - Not observed
    - Mild aggression
    - Moderate aggression
    - Serious aggression
    - High aggression

cbarq_aggression_1	cbarq_aggression_2	cbarq_aggression_3	cbarq_aggression_4	cbarq_aggression_5	cbarq_aggression_6
NA	No aggression	Mild aggression	Moderate aggression	NA	No aggression
NA	NA	NA	NA	NA	NA
No aggression	No aggression	No aggression	No aggression	No aggression	No aggression
No aggression	No aggression	No aggression	No aggression	No aggression	No aggression
NA	NA	NA	NA	NA	NA
No aggression	Mild aggression	NA	Moderate aggression	No aggression	No aggression
No aggression	No aggression	No aggression	No aggression	No aggression	No aggression
No aggression	No aggression	No aggression	No aggression	No aggression	No aggression

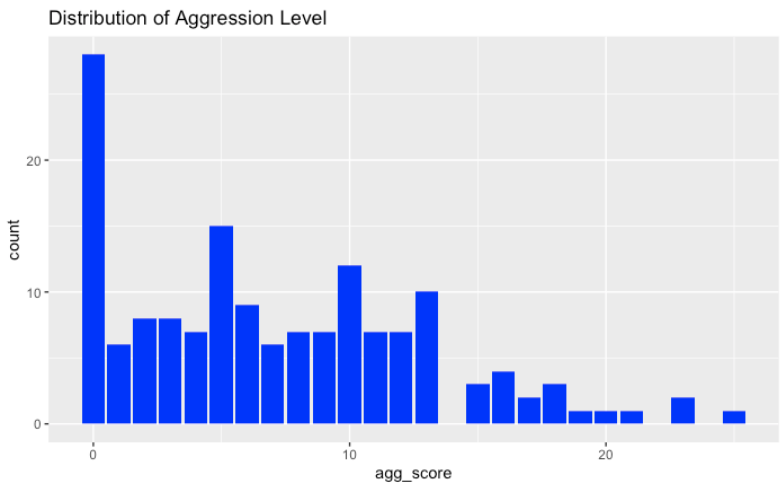
  - The aggression score was computed two ways:
    1. Any scores moderate and above were counted as 1 and total was tallied (ie. Max score of 27)
    2. Scores were ranked 0-3 with 3 being severe and total was tallied (ie. Max score of 3*27 = 81) **Adjusted**

OVERVIEW – INDEPENDENT VARIABLES

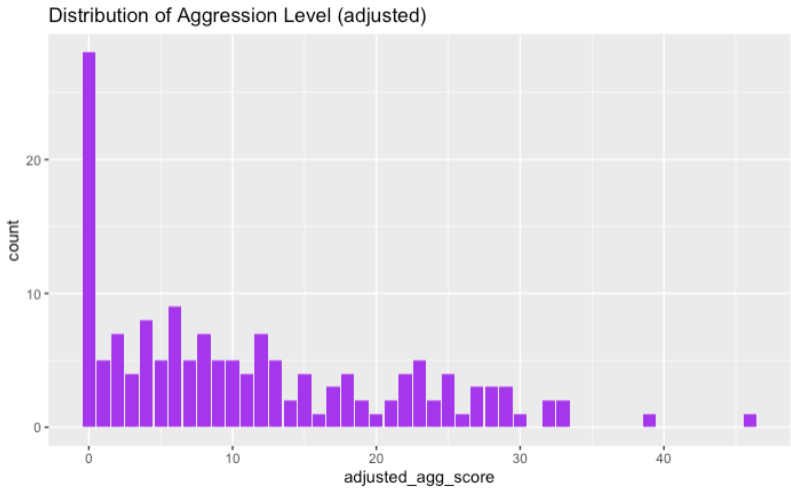
- Environment
- Owned Status (private vs. group housing)
- Sex
- Age
- Desexed
- Pure Bred
- Number of Household Dogs (1, 2… 6+)
- Origin (breeder, rescue, etc.)
- Guardian Gender

Research Question: Which variables can predict dog aggression?

SUMMARY STATISTICS – DEPENDENT VARIABLE

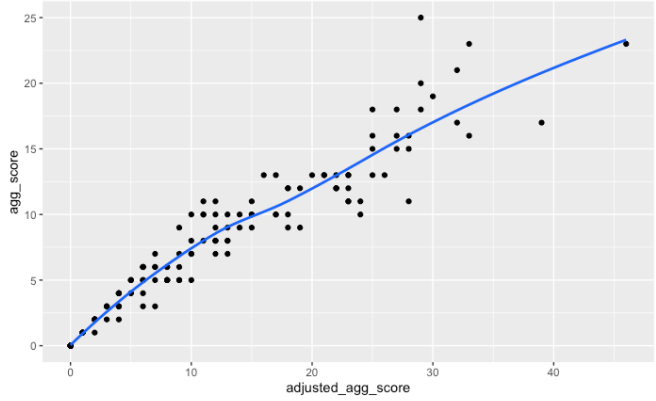



```
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000   2.000   6.000   7.116  11.000  25.000
```



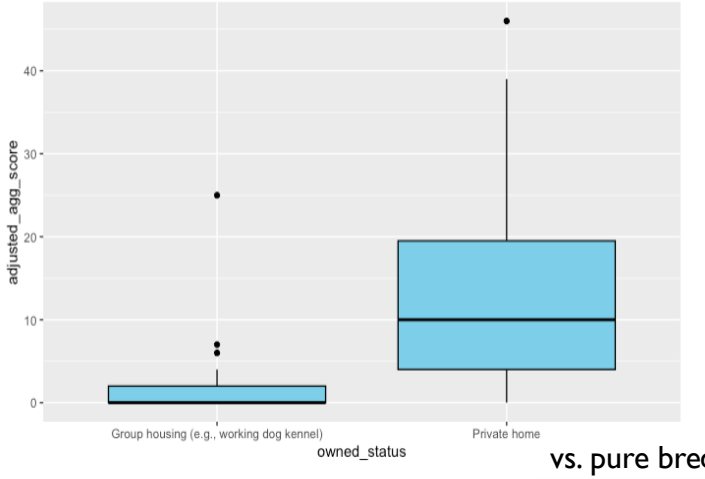

```
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.00    2.00    8.00   11.11   18.00   46.00
```

Aggression Score vs. Adjusted Aggression Scores

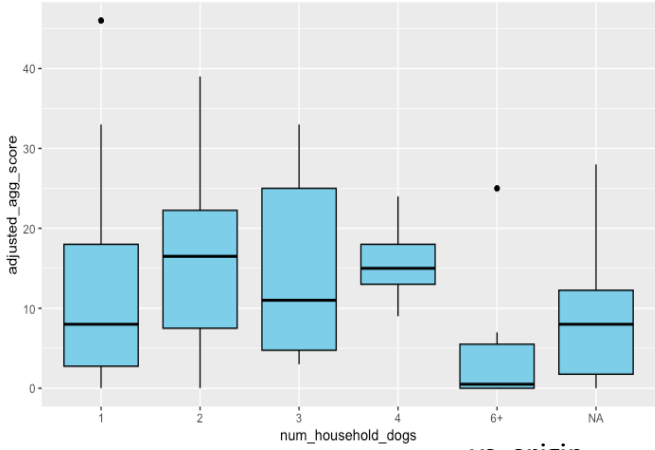


SUMMARY STATISTICS – INDEPENDENT VARIABLES

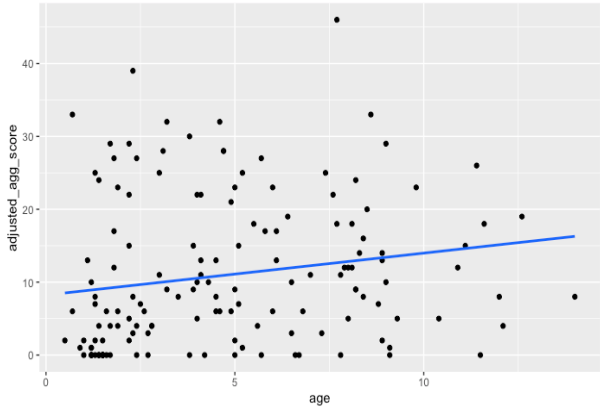
Adjusted aggression vs. Owned Status



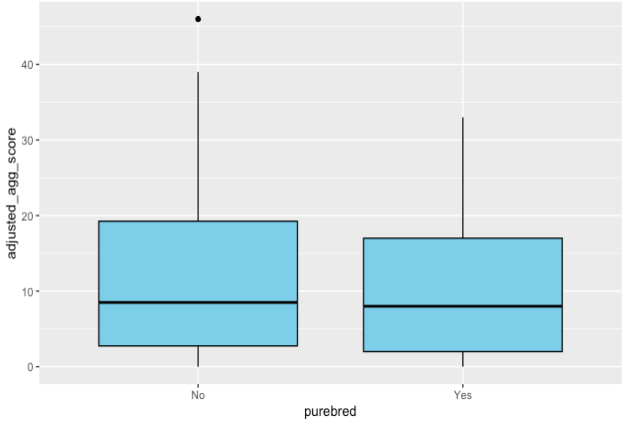
vs. number of household dogs



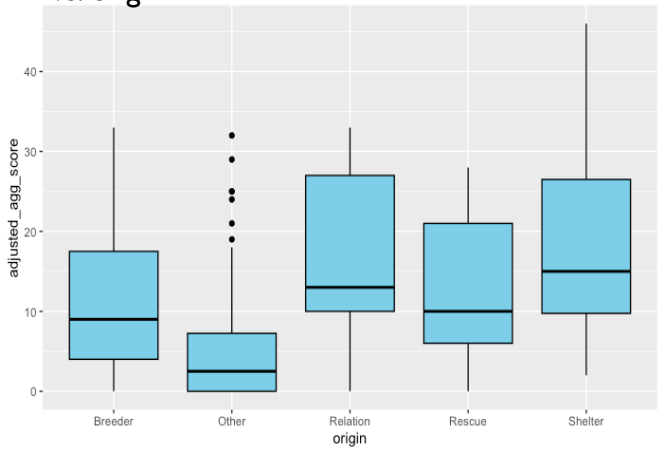
vs age



vs. pure bred



vs. origin



STATISTICAL OUTPUTS – REGRESSION RESULTS

1. Initial (all variable) Result

```
Call:
lm(formula = adjusted_agg_score ~ environment + owned_status +
    sex + age + desexed + purebred + num_household_dogs + origin +
    guardian_gender, data = manydogs_2024_clean_v2)

Residuals:
    Min      1Q  Median      3Q     Max
-16.593  -6.493  -0.987   6.751  23.890

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)               2.27593    5.44773   0.418   0.6770
environmentSuburban       1.87112    3.34413   0.560   0.5770
environmentUrban          1.80776    2.90341   0.623   0.5349
owned_statusPrivate home 12.04326    4.59891   2.619   0.0102 *
sexMale                   0.47972    1.95043   0.246   0.8062
age                      -0.05873    0.34334  -0.171   0.8645
desexedYes               -2.03395    2.43133  -0.837   0.4048
purebredYes              -4.02274    2.34586  -1.715   0.0894 .
num_household_dogs2       4.66911    2.69399   1.733   0.0861 .
num_household_dogs3       3.20831    3.30601   0.970   0.3341
num_household_dogs4       4.37871    4.94572   0.885   0.3781
num_household_dogs6+      4.89445    4.93937   0.991   0.3241
originOther              -2.17000    3.05493  -0.710   0.4791
originRelation            8.72049    4.44841   1.960   0.0527 .
originRescue             -1.34355    3.18286  -0.422   0.6738
originShelter             9.33370    4.48693   2.080   0.0400 *
guardian_genderMale      -1.97344    3.26463  -0.604   0.5469
guardian_genderOther      6.51462    5.37207   1.213   0.2281
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.702 on 101 degrees of freedom
  (36 observations deleted due to missingness)
Multiple R-squared:  0.297,     Adjusted R-squared:  0.1787
F-statistic:  2.51 on 17 and 101 DF,  p-value: 0.002367
```

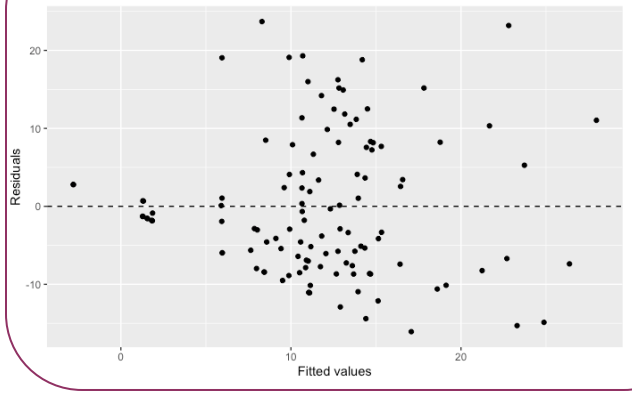
2. Final result using backwards selection

```
Call:
lm(formula = adjusted_agg_score ~ owned_status + purebred + age +
    num_household_dogs + origin, data = manydogs_2024_clean_v2)

Residuals:
    Min      1Q  Median      3Q     Max
-16.077  -7.138  -1.787   6.960  23.702

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)                3.7972     4.6627   0.814  0.41724
owned_statusPrivate home  11.5787     4.3351   2.671  0.00875 **
purebredYes               -4.0971     2.2682  -1.806  0.07368 .
age                       -0.1538     0.3232  -0.476  0.63520
num_household_dogs2        4.3269     2.4816   1.744  0.08410 .
num_household_dogs3        3.0231     3.2159   0.940  0.34931
num_household_dogs4        3.4658     4.5120   0.768  0.44410
num_household_dogs6+       4.6222     4.4195   1.046  0.29799
originOther               -2.2729     2.9550  -0.769  0.44349
originRelation             7.8655     4.2471   1.852  0.06678 .
originRescue              -1.8257     2.8874  -0.632  0.52854
originShelter              8.6146     4.2561   2.024  0.04546 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.545 on 107 degrees of freedom
  (36 observations deleted due to missingness)
Multiple R-squared:  0.2791,    Adjusted R-squared:  0.205
F-statistic: 3.766 on 11 and 107 DF,  p-value: 0.0001432
```

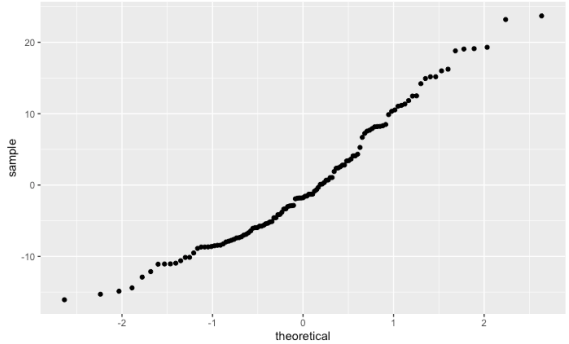
Age ~0.63

STATISTICAL OUTPUTS – CHECKING ASSUMPTIONS

#Plot 1 Residuals vs. Fitted Values

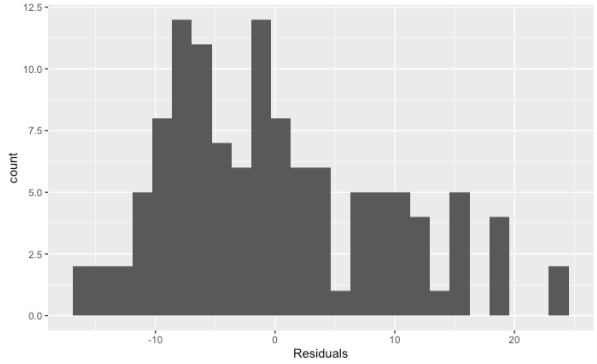


Plot 3 Normal prob plot of residuals



^ plot looks overall randomly scattered though looks slightly wider on right. *Impact of categorical data?*

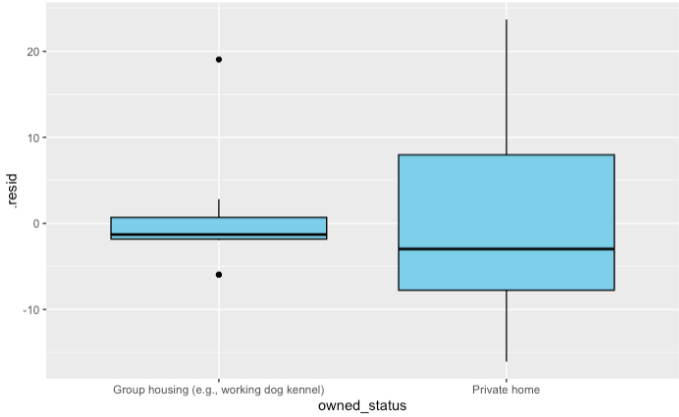
Plot 2 Distribution of residuals



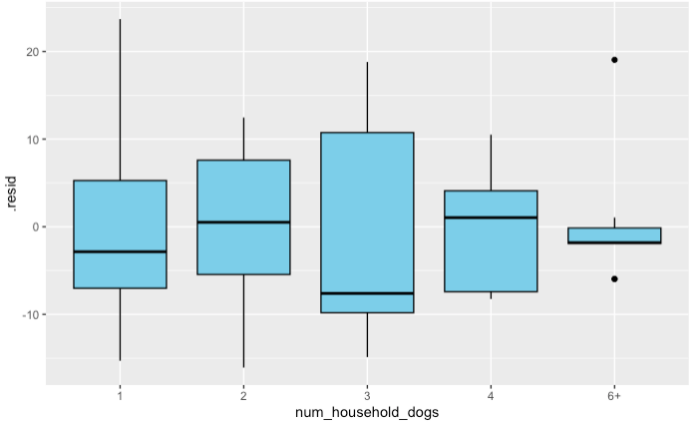
< ^ residuals appear normally distributed and nearly normal.

STATISTICAL OUTPUTS – CHECKING ASSUMPTIONS

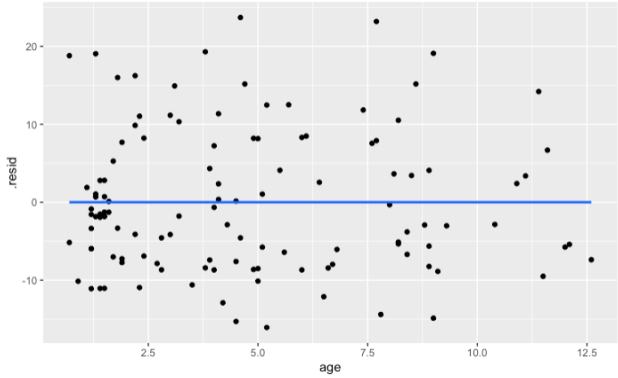
Residuals vs. Owned Status



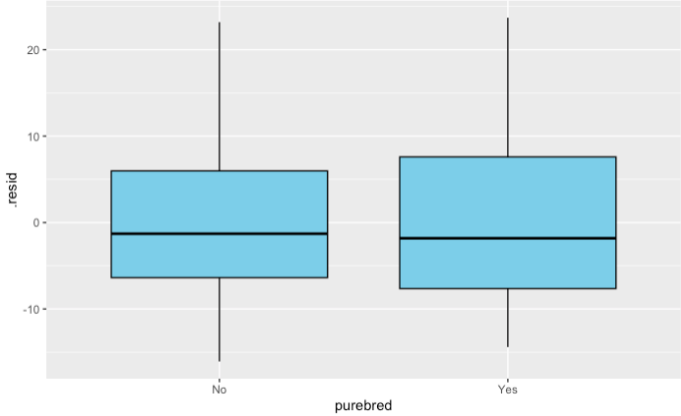
vs. number of household dogs



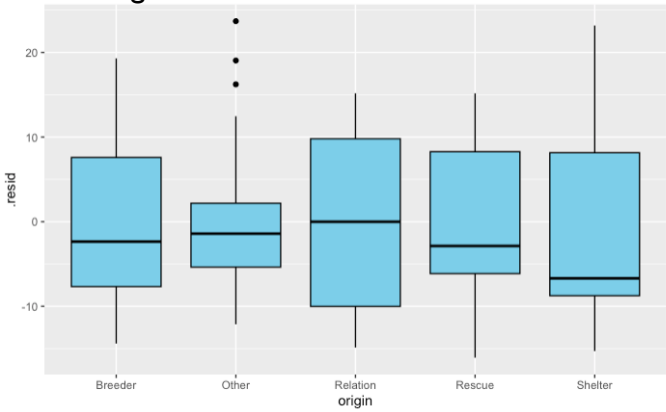
vs. age



vs. pure bred



vs. origin



STATISTICAL OUTPUTS – ANOVA TABLES

- Null hypothesis that there are no significant difference in mean across groups.
- Review of ANOVA tables showed as expected – Number of household dogs, owned status and origin had significant P values to reject the null hypothesis in favor of the alternative that at least one group mean differs.
- Age was found to be significant which prompted for inclusion into regression.

Number of household dogs

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
num_household_dogs	4	950	237.4	2.187	0.0745	.
Residuals	118	12808	108.5			

Owned Status

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
owned_status	1	1742	1741.9	18.82	2.59e-05	***
Residuals	153	14157	92.5			

Origin

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
origin	4	1809	452.1	4.813	0.00111	**
Residuals	150	14091	93.9			

Age

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
age	1	509	509.3	4.982	0.0271	*
Residuals	148	15130	102.2			

Purebred

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
purebred	1	40	40.14	0.387	0.535
Residuals	153	15859	103.65		

STATISTICAL OUTPUTS – ANOVA TABLES

- Interestingly no interaction effects were found between variables

Environment & household dogs

```
Df Sum Sq Mean Sq F value Pr(>F)
environment                 2    165   82.44   0.743  0.478
num_household_dogs          4    870  217.53   1.959  0.106
environment:num_household_dogs  6    510   85.05   0.766  0.598
Residuals                 110  12213  111.02
```

Owned status & household dogs

```
32 observations deleted due to missingness
                        Df Sum Sq Mean Sq F value Pr(>F)
owned_status                1   1953  1953.2  19.788  2e-05 ***
num_household_dogs          4    326    81.6   0.827  0.511
owned_status:num_household_dogs  1     28    28.4   0.288  0.593
Residuals                 116  11450    98.7
```

- Accept null hypothesis there is no mean diff between urban, suburban and rural

Environment

```
Df Sum Sq Mean Sq F value Pr(>F)
environment   2     72   36.13   0.347  0.707
Residuals   152  15827  104.12
```

CONCLUSION – FINAL REGRESSION MODEL

Aggression score equation = 3.7972 *(intercept)*

1 + 11.5787 * **owned_status** (where private home = 1, group housing = 0)

2 - 4.0971 * **purebred** (where yes = 1, otherwise 0)

3 - 0.1538 * **age**

4 + 4.3269 * **num_household_dogs**(2)

+ 3.0231 * num_household_dogs(3)

+ 3.4658 * num_household_dogs(4)

+ 4.6222 * num_household_dogs(6+)

5 - 2.2729 * **origin**(Other)

+ 7.8655 * origin(Relation)

- 1.8257 * origin(Rescue)

+ 8.6146* origin(Shelter)

Dogs in private home settings that have more than 1 dog, non purebreds and are either from "relations" or shelter are more likely to be aggressive.

Furthermore, aggression declines slightly with age.

Note: num_household_dogs 1 is default; for origin, breeder is the default

CONCLUSION – FINAL THOUGHTS & TAKEAWAYS

- Data clean up takes longer than one would think! (and it's a shame to waste collected data that have one or two NAs)

- Dogs in private home settings that have more than one dog, non purebreds and either from "relations" or shelter are more likely to be aggressive. Furthermore, aggression declines slightly with age.
  - Some of the results make intuitive sense (e.g., aggression declines slightly with age, shelter dogs more likely to be aggressive)
  - Interestingly, I learned the diff between "shelters" and "rescues" and shelters add to aggression score while rescue decreases it. (Could be explained that shelters can house a lot of dogs and be overrun)
  - With more dogs, it's possible that the chance of reaction capturing is higher
  - Some ambiguity with survey answers (e.g. how is urban, suburban vs. rural defined?)

- With adjusted R^2 of ~0.2, there are a lot of other factors the current model does not capture.
  - Would more continuous quantitative variables result in better model?

- Adjusted aggression score had slight higher explanatory results, perhaps because the results can be better defined with more range.

- Initial curiosity if "environment" (urban vs. rural) impacts aggression was answered and more.

THANK YOU!